cadence

JasperGold® - Superlint Violation/Property Report

module *riscv_scalar_cryptography_proscessor* created date *Feb 16 07:54:40 IST 2023*

VIOLATIONS SUMMARY

Domain: LINT

#	Category	Error	Warning	Info	Waived	Total
1	CODINGSTYLE	0	0	0	149	149
2	STRUCTURAL	0	0	0	129	129
3	SYNTHESIS	0	0	0	6	6
4	Total	0	0	0	284	284

CODINGSTYLE

#	Severity	Tag	Message	Source Location	Instance-Module	Grouping Status
1	Waived (ARY_MS_DRNG waived because mixed packed/unpacked range ordering is intentional and legal Verilog; no functional or synthesis impact.)	ARY_MS_DRNG	"Inconsistent ordering of bits in range declarations in design-unit im_mem0 -- should be all Descending ranges"	im_mem0.v :11	*-im_mem0	-
2	Waived (ARY_MS_DRNG waived because mixed packed/unpacked range ordering is intentional and legal Verilog; no functional or synthesis impact.)	ARY_MS_DRNG	"Inconsistent ordering of bits in range declarations in design-unit im_mem1 -- should be all Descending ranges"	im_mem1.v :12	*-im_mem1	-
3	Waived (ARY_MS_DRNG waived because mixed packed/unpacked range ordering is intentional and legal Verilog; no functional or synthesis impact.)	ARY_MS_DRNG	"Inconsistent ordering of bits in range declarations in design-unit im_mem2 -- should be all Descending ranges"	im_mem2.v :11	*-im_mem2	-
4	Waived (ARY_MS_DRNG waived because mixed packed/unpacked range ordering is intentional and legal Verilog; no functional or synthesis impact.)	ARY_MS_DRNG	"Inconsistent ordering of bits in range declarations in design-unit im_mem3 -- should be all Descending ranges"	im_mem3.v :9	*-im_mem3	-
5	Waived (ARY_MS_DRNG waived because mixed packed/unpacked range ordering is intentional and legal Verilog; no functional or synthesis impact.)	ARY_MS_DRNG	"Inconsistent ordering of bits in range declarations in design-unit gpr -- should be all Descending ranges"	register_file.v :12	*-gpr	-
6	Waived (ARY_MS_DRNG waived because mixed packed/unpacked range ordering is intentional and legal Verilog; no functional or synthesis impact.)	ARY_MS_DRNG	"Inconsistent ordering of bits in range declarations in design-unit data_memory -- should be all Descending ranges"	data_memory.v :9	*-data_memory	-
7	Waived (Unequal operand width warning waived because operands are intentionally widened for PC increment logic and truncation to	ASG_MS_RPAD	"Unequal length operands in assignment in design-unit/block 'pc_add'. Length of RHS is less than LHS. LHS 'pc_in_wide' (unsigned) - 33 bit(s), RHS '(pc_next + {{(XLEN - 3){1'b0}},3'b100})' (unsigned) - 32 bit(s). RHS will be padded by 1 bit(s)."	pc_adder.v :18	*-pc_add	-

	XLEN bits is architecturally correct; no functional or synthesis impact.)					
8	Waived (Unequal operand width warning waived because operands are intentionally widened for PC increment logic and truncation to XLEN bits is architecturally correct; no functional or synthesis impact.)	ASG_MS_RPAD	"Unequal length operands in assignment in design-unit/block 'sha256_round'. Length of RHS is less than LHS. LHS 'T1_wide' (unsigned) - 35 bit(s), RHS '((((h + S1(e)) + Ch(e,f,g)) + k) + w)' (unsigned) - 32 bit(s). RHS will be padded by 3 bit(s)."	sha256_round.v :100	*-sha256_round	-
9	Waived (Unequal operand width warning waived because operands are intentionally widened for PC increment logic and truncation to XLEN bits is architecturally correct; no functional or synthesis impact.)	ASG_MS_RPAD	"Unequal length operands in assignment in design-unit/block 'sha256_round'. Length of RHS is less than LHS. LHS 'T2_wide' (unsigned) - 34 bit(s), RHS '(S0(a) + Maj(a,b,c))' (unsigned) - 32 bit(s). RHS will be padded by 2 bit(s)."	sha256_round.v :101	*-sha256_round	-
10	Waived (Unequal operand width warning waived because operands are intentionally widened for PC increment logic and truncation to XLEN bits is architecturally correct; no functional or synthesis impact.)	ASG_MS_RPAD	"Unequal length operands in assignment in design-unit/block 'sha256_round'. Length of RHS is less than LHS. LHS 'ao_wide' (unsigned) - 33 bit(s), RHS '(T1 + T2)' (unsigned) - 32 bit(s). RHS will be padded by 1 bit(s)."	sha256_round.v :106	*-sha256_round	-
11	Waived (Unequal operand width warning waived because operands are intentionally widened for PC increment logic and truncation to XLEN bits is architecturally correct; no functional or synthesis impact.)	ASG_MS_RPAD	"Unequal length operands in assignment in design-unit/block 'sha256_round'. Length of RHS is less than LHS. LHS 'eo_wide' (unsigned) - 33 bit(s), RHS '(d + T1)' (unsigned) - 32 bit(s). RHS will be padded by 1 bit(s)."	sha256_round.v :107	*-sha256_round	-
12	Waived (Unequal operand width warning waived because operands are intentionally widened for PC increment logic and truncation to XLEN bits is architecturally correct; no functional or synthesis impact.)	ASG_MS_RPAD	"Unequal length operands in assignment in design-unit/block 'sha256_msg_datapath'. Length of RHS is less than LHS. LHS 'w0_wide' (unsigned) - 34 bit(s), RHS '(((SIG1(bufrd_t2_in) + bufrd_t7_in) + SIG0(bufrd_t15_in)) + bufrd_t16_in)' (unsigned) - 32 bit(s). RHS will be padded by 2 bit(s)."	sha256_msg_datapath.v :143	*-sha256_msg_datapath	-
13	Waived (Unequal operand width warning waived because operands are intentionally widened for PC increment logic and truncation to XLEN bits is architecturally correct; no functional or synthesis impact.)	ASG_MS_RPAD	"Unequal length operands in assignment in design-unit/block 'sha256_msg_datapath'. Length of RHS is less than LHS. LHS 'w1_wide' (unsigned) - 34 bit(s), RHS '(((SIG1(bufrd_t1_in) + bufrd_t6_in) + SIG0(bufrd_t14_in)) + bufrd_t15_in)' (unsigned) - 32 bit(s). RHS will be padded by 2 bit(s)."	sha256_msg_datapath.v :146	*-sha256_msg_datapath	-

14	Waived (Unequal operand width warning waived because operands are intentionally widened for PC increment logic and truncation to XLEN bits is architecturally correct; no functional or synthesis impact.)	ASG_MS_RPAD	"Unequal length operands in assignment in design-unit/block 'sha256_msg_datapath'. Length of RHS is less than LHS. LHS 'w2_wide' (unsigned) - 34 bit(s), RHS '(((SIG1(w0_int) + bufrd_t5_in) + SIG0(bufrd_t13_in)) + bufrd_t14_in)' (unsigned) - 32 bit(s). RHS will be padded by 2 bit(s)."	sha256_msg_datapath.v :149	*-sha256_msg_datapath	-
15	Waived (Unequal operand width warning waived because operands are intentionally widened for PC increment logic and truncation to XLEN bits is architecturally correct; no functional or synthesis impact.)	ASG_MS_RPAD	"Unequal length operands in assignment in design-unit/block 'sha256_msg_datapath'. Length of RHS is less than LHS. LHS 'w3_wide' (unsigned) - 34 bit(s), RHS '(((SIG1(w1_int) + bufrd_t4_in) + SIG0(bufrd_t12_in)) + bufrd_t13_in)' (unsigned) - 32 bit(s). RHS will be padded by 2 bit(s)."	sha256_msg_datapath.v :152	*-sha256_msg_datapath	-
16	Waived (ASG_NR_POVF waived because the widened left-shift in rotate logic may trigger conservative overflow detection, but truncation is intentional and functionally correct)	ASG_NR_POVF	"Potential overflow in assignment in design-unit/block 'bit_ror'. LHS 'lshift_wide' (unsigned) is of 33 bit(s), RHS '({1'b0,rs1} << (XLEN - shamt))' (unsigned) can be of 64 bit(s) "	bit_ror.v :60	*-bit_ror	-
17	Waived (CAS_NR_DEFX waived because the default case is intentionally fully specified by prior assignments and outputs are not required to be driven to X; no functional or synthesis impact.)	CAS_NR_DEFX	"Signal 'bitmanipulation_out' assigned in case statement is not assigned to X in default case"	bitmanipulation.v :55	*-bitmanipulation	-
18	Waived (CAS_NR_DEFX waived because the default case is intentionally fully specified by prior assignments and outputs are not required to be driven to X; no functional or synthesis impact.)	CAS_NR_DEFX	"Signal 'carry_less_mul_out' assigned in case statement is not assigned to X in default case"	carry_less_engine.v :34	*-carry_less_mul_top	-
19	Waived (CAS_NR_DEFX waived because the default case is intentionally fully specified by prior assignments and outputs are not required to be driven to X; no functional or synthesis impact.)	CAS_NR_DEFX	"Signal 'out_byte' assigned in case statement is not assigned to X in default case"	sbox_values.v :7	*-aes_sbox	-
20	Waived (CAS_NR_DEFX waived because the default case is intentionally fully specified by prior assignments and outputs are not required to be driven to X; no functional or	CAS_NR_DEFX	"Signal 'byte_in' assigned in case statement is not assigned to X in default case"	aes32esi.v :18	*-aes32esi	-

	synthesis impact.)					
21	Waived (CAS_NR_DEFX waived because the default case is intentionally fully specified by prior assignments and outputs are not required to be driven to X; no functional or synthesis impact.)	CAS_NR_DEFX	"Signal 'roli' assigned in case statement is not assigned to X in default case"	aes32esi.v :40	*-aes32esi	-
22	Waived (CAS_NR_DEFX waived because the default case is intentionally fully specified by prior assignments and outputs are not required to be driven to X; no functional or synthesis impact.)	CAS_NR_DEFX	"Signal 'byte_in' assigned in case statement is not assigned to X in default case"	aes32esmi.v :19	*-aes32esmi	-
23	Waived (CAS_NR_DEFX waived because the default case is intentionally fully specified by prior assignments and outputs are not required to be driven to X; no functional or synthesis impact.)	CAS_NR_DEFX	"Signal 'roli' assigned in case statement is not assigned to X in default case"	aes32esmi.v :47	*-aes32esmi	-
24	Waived (CAS_NR_DEFX waived because the default case is intentionally fully specified by prior assignments and outputs are not required to be driven to X; no functional or synthesis impact.)	CAS_NR_DEFX	"Signal 'aes_out' assigned in case statement is not assigned to X in default case"	aes_engine.v :25	*-aes_engine	-
25	Waived (CAS_NR_DEFX waived because the default case is intentionally fully specified by prior assignments and outputs are not required to be driven to X; no functional or synthesis impact.)	CAS_NR_DEFX	"Signal 'sha_engine_out' assigned in case statement is not assigned to X in default case"	sha_engine.v :57	*-sha_engine	-
26	Waived (IDX_NR_DTTY waived because the address used as a memory index is driven from registered control logic and is guaranteed to be a valid 2-state value; no functional or synthesis impact.)	IDX_NR_DTTY	"Variable 'pc_addr' used as index in expression 'mem[pc_addr]' should be 2-state data type"	im_mem0.v :18	*-im_mem0	-
27	Waived (IDX_NR_DTTY waived because the address used as a memory index is driven from registered control logic and is guaranteed to be a valid 2-state value; no	IDX_NR_DTTY	"Variable 'pc_addr' used as index in expression 'mem[pc_addr]' should be 2-state data type"	im_mem0.v :22	*-im_mem0	-

	functional or synthesis impact.)					
28	Waived (IDX_NR_DTTY waived because the address used as a memory index is driven from registered control logic and is guaranteed to be a valid 2-state value; no functional or synthesis impact.)	IDX_NR_DTTY	"Variable 'pc_addr' used as index in expression 'mem[pc_addr]' should be 2-state data type"	im_mem1.v :21	*-im_mem1	-
29	Waived (IDX_NR_DTTY waived because the address used as a memory index is driven from registered control logic and is guaranteed to be a valid 2-state value; no functional or synthesis impact.)	IDX_NR_DTTY	"Variable 'pc_addr' used as index in expression 'mem[pc_addr]' should be 2-state data type"	im_mem1.v :25	*-im_mem1	-
30	Waived (IDX_NR_DTTY waived because the address used as a memory index is driven from registered control logic and is guaranteed to be a valid 2-state value; no functional or synthesis impact.)	IDX_NR_DTTY	"Variable 'pc_addr' used as index in expression 'mem[pc_addr]' should be 2-state data type"	im_mem2.v :18	*-im_mem2	-
31	Waived (IDX_NR_DTTY waived because the address used as a memory index is driven from registered control logic and is guaranteed to be a valid 2-state value; no functional or synthesis impact.)	IDX_NR_DTTY	"Variable 'pc_addr' used as index in expression 'mem[pc_addr]' should be 2-state data type"	im_mem2.v :22	*-im_mem2	-
32	Waived (IDX_NR_DTTY waived because the address used as a memory index is driven from registered control logic and is guaranteed to be a valid 2-state value; no functional or synthesis impact.)	IDX_NR_DTTY	"Variable 'pc_addr' used as index in expression 'mem[pc_addr]' should be 2-state data type"	im_mem3.v :16	*-im_mem3	-
33	Waived (IDX_NR_DTTY waived because the address used as a memory index is driven from registered control logic and is guaranteed to be a valid 2-state value; no functional or synthesis impact.)	IDX_NR_DTTY	"Variable 'pc_addr' used as index in expression 'mem[pc_addr]' should be 2-state data type"	im_mem3.v :20	*-im_mem3	-
34	Waived (IDX_NR_DTTY waived because the address used as a memory index is driven from registered control logic and is guaranteed to be a valid 2-state	IDX_NR_DTTY	"Variable 'rd_addr' used as index in expression 'mem[rd_addr]' should be 2-state data type"	register_file.v :44	*-gpr	-

	value; no functional or synthesis impact.)					
35	Waived (IDX_NR_DTTY waived because the address used as a memory index is driven from registered control logic and is guaranteed to be a valid 2-state value; no functional or synthesis impact.)	IDX_NR_DTTY	"Variable 'rd_addr' used as index in expression 'mem[rd_addr]' should be 2-state data type"	register_file.v :50	*-gpr	-
36	Waived (IDX_NR_DTTY waived because the address used as a memory index is driven from registered control logic and is guaranteed to be a valid 2-state value; no functional or synthesis impact.)	IDX_NR_DTTY	"Variable 'addr' used as index in expression 'mem[addr]' should be 2-state data type"	data_memory.v :20	*-data_memory	-
37	Waived (IDX_NR_DTTY waived because the address used as a memory index is driven from registered control logic and is guaranteed to be a valid 2-state value; no functional or synthesis impact.)	IDX_NR_DTTY	"Variable 'addr' used as index in expression 'mem[addr]' should be 2-state data type"	data_memory.v :24	*-data_memory	-
38	Waived (Unused input bits are intentional due to instruction semantics or address slicing; no functional or synthesis impact.)	INP_NO_USED	"The input port 'inst_in[19:0]' declared in the module 'riscv_control_unit' is unused"	riscv_control_unit.v :6	*-riscv_control_unit	-
39	Waived (Unused input bits are intentional due to instruction semantics or address slicing; no functional or synthesis impact.)	INP_NO_USED	"The input port 'rs2[31:5]' declared in the module 'bit_rol' is unused"	bit_rol.v :24	*-bit_rol	-
40	Waived (Unused input bits are intentional due to instruction semantics or address slicing; no functional or synthesis impact.)	INP_NO_USED	"The input port 'rs2[31:5]' declared in the module 'bit_ror' is unused"	bit_ror.v :43	*-bit_ror	-
41	Waived (Unused input bits are intentional due to instruction semantics or address slicing; no functional or synthesis impact.)	INP_NO_USED	"The input port 'rs1[31:16]' declared in the module 'bit_pack' is unused"	bit_pack.v :2	*-bit_pack	-
42	Waived (Unused input bits are intentional due to instruction semantics or address slicing; no functional or synthesis impact.)	INP_NO_USED	"The input port 'rs2[15:0]' declared in the module 'bit_pack' is unused"	bit_pack.v :3	*-bit_pack	-
43	Waived (Unused input bits are intentional due to instruction semantics or	INP_NO_USED	"The input port 'rs1[31:8]' declared in the module 'bit_packh' is unused"	bit_packh.v :2	*-bit_packh	-

	address slicing; no functional or synthesis impact.)					
44	Waived (Unused input bits are intentional due to instruction semantics or address slicing; no functional or synthesis impact.)	INP_NO_USED	"The input port 'rs2[31:8]' declared in the module 'bit_packh' is unused"	bit_packh.v :3	*-bit_packh	-
45	Waived (Unused input bits are intentional due to instruction semantics or address slicing; no functional or synthesis impact.)	INP_NO_USED	"The input port 'addr[1:0]' declared in the module 'data_memory_top' is unused"	data_memory_top.v :1	*-data_memory_top	-
46	Waived (Unused input bits are intentional due to instruction semantics or address slicing; no functional or synthesis impact.)	INP_NO_USED	"The input port 'addr[31:10]' declared in the module 'data_memory_top' is unused"	data_memory_top.v :1	*-data_memory_top	-
47	Waived (OPR_NR_UEAS waived because subtraction between XLEN and shamt is intentional for rotate logic and width difference has no functional or synthesis impact.)	OPR_NR_UEAS	"Unequal length operand in arithmetic operator subtraction in design-unit/block bit_rol. LHS operand 'XLEN' is 32 bits, RHS operand 'shamt' is 5 bits"	bit_rol.v :35	*-bit_rol	-
48	Waived (OPR_NR_UEAS waived because subtraction between XLEN and shamt is intentional for rotate logic and width difference has no functional or synthesis impact.)	OPR_NR_UEAS	"Unequal length operand in arithmetic operator subtraction in design-unit/block bit_ror. LHS operand 'XLEN' is 32 bits, RHS operand 'shamt' is 5 bits"	bit_ror.v :60	*-bit_ror	-
49	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'pc_counter.pc_adder.pc_in' is assigned asynchronously"	pc_adder.v :13	pc_counter.pc_adder-pc_add	-
50	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'instruction.im_memo0.byte_0_out' is assigned asynchronously"	im_mem0.v :7	instruction.im_memo0-im_mem0	-
51	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'instruction.im_memo1.byte_1_out' is assigned asynchronously"	im_mem1.v :7	instruction.im_memo1-im_mem1	-

52	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'instruction.im_memo2.byte_2_out' is assigned asynchronously"	im_mem2.v :7	instruction.im_memo2-im_mem2	-
53	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'instruction.im_memo3.byte_3_out' is assigned asynchronously"	im_mem3.v :6	instruction.im_memo3-im_mem3	-
54	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'instruction.inst_out_c' is assigned asynchronously"	im_mem_top.v :6	instruction-instruction_memory_top	-
55	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'riscv_contro_unit.alu_opcode' is assigned asynchronously"	riscv_control_unit.v :11	riscv_contro_unit-riscv_control_unit	-
56	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'riscv_contro_unit.alu_sel_mux' is assigned asynchronously"	riscv_control_unit.v :12	riscv_contro_unit-riscv_control_unit	-
57	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'riscv_contro_unit.byte_sel' is assigned asynchronously"	riscv_control_unit.v :13	riscv_contro_unit-riscv_control_unit	-
58	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'riscv_contro_unit.bytes' is assigned asynchronously"	riscv_control_unit.v :14	riscv_contro_unit-riscv_control_unit	-
59	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure	OTP_NR_ASYA	"Output port 'riscv_contro_unit.data_sel_mux' is assigned asynchronously"	riscv_control_unit.v :10	riscv_contro_unit-riscv_control_unit	-

	combinational logic and is not required to be registered; no functional or synthesis impact.)					
60	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'riscv_contro_unit.data_write_en' is assigned asynchronously"	riscv_control_unit.v :9	riscv_contro_unit-riscv_control_unit	-
61	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'riscv_contro_unit.imm_sel' is assigned asynchronously"	riscv_control_unit.v :8	riscv_contro_unit-riscv_control_unit	-
62	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'riscv_contro_unit.reg_write' is assigned asynchronously"	riscv_control_unit.v :7	riscv_contro_unit-riscv_control_unit	-
63	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'riscv_contro_unit.sign_extend' is assigned asynchronously"	riscv_control_unit.v :15	riscv_contro_unit-riscv_control_unit	-
64	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'register_file.rs1_data' is assigned asynchronously"	register_file.v :8	register_file-gpr	-
65	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'register_file.rs2_data' is assigned asynchronously"	register_file.v :9	register_file-gpr	-
66	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.	OTP_NR_ASYA	"Output port 'signextend.imm_out' is assigned asynchronously"	sign_extend.v :3	signextend-sign_extend	-

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67	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'mux21.mux_out' is assigned asynchronously"	mux_21.v :4	mux21-mux	-
68	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.bit_manipulation_engine.bitrol.rol_result' is assigned asynchronously"	bit_rol.v :25	crypto_alu.bit_manipulation_engine.bitrol-bit_rol	-
69	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.bit_manipulation_engine.bitror.ror_result' is assigned asynchronously"	bit_ror.v :44	crypto_alu.bit_manipulation_engine.bitror-bit_ror	-
70	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.bit_manipulation_engine.bitandn.andn_result' is assigned asynchronously"	bit_andn.v :4	crypto_alu.bit_manipulation_engine.bitandn-bit_andn	-
71	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.bit_manipulation_engine.bitorn.orn_result' is assigned asynchronously"	bit_orn.v :4	crypto_alu.bit_manipulation_engine.bitorn-bit_orn	-
72	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.bit_manipulation_engine.bitxnor.xnor_result' is assigned asynchronously"	bit_xnor.v :4	crypto_alu.bit_manipulation_engine.bitxnor-bit_xnor	-
73	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.bit_manipulation_engine.bitpack.pack_result[31:16]' is assigned asynchronously"	bit_pack.v :4	crypto_alu.bit_manipulation_engine.bitpack-bit_pack	-
74	Waived (OTP_NR_ASYA waived because this output is intentionally	OTP_NR_ASYA	"Output port 'crypto_alu.bit_manipulation_engine.bitpackh.packh_result[31:8]' is assigned asynchronously"	bit_packh.v :4	crypto_alu.bit_manipulation_engine.bitpackh-bit_packh	-

	driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)					
75	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.bit_manipulation_engine.bitmanipulation_out' is assigned asynchronously"	bitmanipulation.v :5	crypto_alu.bit_manipulation_engine-bitmanipulation	-
76	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.carry_less_mul_top.carry_less_mul_lower_32.carry_less_out' is assigned asynchronously"	carry_less_multiplication.v :85	crypto_alu.carry_less_mul_top.carry_less_mul_lower_32-carry_less_mul	-
77	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.carry_less_mul_top.carry_less_mul_out' is assigned asynchronously"	carry_less_engine.v :5	crypto_alu.carry_less_mul_top-carry_less_mul_top	-
78	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.aes_engine_top.aes32sbox.sbox.out_byte' is assigned asynchronously"	sbox_values.v :3	crypto_alu.aes_engine_top.aes32sbox.sbox-aes_sbox	-
79	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.aes_engine_top.aes32sbox.aes32esi_out' is assigned asynchronously"	aes32esi.v :6	crypto_alu.aes_engine_top.aes32sbox-aes32esi	-
80	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.aes_engine_top.aes32smix.sbox.out_byte' is assigned asynchronously"	sbox_values.v :3	crypto_alu.aes_engine_top.aes32smix.sbox-aes_sbox	-
81	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or	OTP_NR_ASYA	"Output port 'crypto_alu.aes_engine_top.aes32smix.mixcolumn.out_word' is assigned asynchronously"	mix_op.v :64	crypto_alu.aes_engine_top.aes32smix.mixcolumn-mix_op	-

	synthesis impact.)					
82	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.aes_engine_top.aes32smix.aes32esmi_out' is assigned asynchronously"	aes32esmi.v :6	crypto_alu.aes_engine_top.aes32smix-aes32esmi	-
83	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.aes_engine_top.aes_out' is assigned asynchronously"	aes_engine.v :7	crypto_alu.aes_engine_top-aes_engine	-
84	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.sha_engine_top.sha_sum0_sum1.ao' is assigned asynchronously"	sha256_round.v :55	crypto_alu.sha_engine_top.sha_sum0_sum1-sha256_round	-
85	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.sha_engine_top.sha_sum0_sum1.bo' is assigned asynchronously"	sha256_round.v :55	crypto_alu.sha_engine_top.sha_sum0_sum1-sha256_round	-
86	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.sha_engine_top.sha_sum0_sum1.co' is assigned asynchronously"	sha256_round.v :55	crypto_alu.sha_engine_top.sha_sum0_sum1-sha256_round	-
87	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.sha_engine_top.sha_sum0_sum1.d0' is assigned asynchronously"	sha256_round.v :55	crypto_alu.sha_engine_top.sha_sum0_sum1-sha256_round	-
88	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.sha_engine_top.sha_sum0_sum1.eo' is assigned asynchronously"	sha256_round.v :55	crypto_alu.sha_engine_top.sha_sum0_sum1-sha256_round	-
89	Waived (OTP_NR_ASYA waived because this output is	OTP_NR_ASYA	"Output port 'crypto_alu.sha_engine_top.sha_sum0_sum1.fo' is assigned asynchronously"	sha256_round.v :55	crypto_alu.sha_engine_top.sha_sum0_sum1-sha256_round	-

	intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)					
90	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.sha_engine_top.sha_sum0_sum1.go' is assigned asynchronously"	sha256_round.v :55	crypto_alu.sha_engine_top.sha_sum0_sum1-sha256_round	-
91	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.sha_engine_top.sha_sum0_sum1.ho' is assigned asynchronously"	sha256_round.v :55	crypto_alu.sha_engine_top.sha_sum0_sum1-sha256_round	-
92	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.sha_engine_top.sha_sum0_sum1.sum0' is assigned asynchronously"	sha256_round.v :56	crypto_alu.sha_engine_top.sha_sum0_sum1-sha256_round	-
93	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.sha_engine_top.sha_sum0_sum1.sum1' is assigned asynchronously"	sha256_round.v :57	crypto_alu.sha_engine_top.sha_sum0_sum1-sha256_round	-
94	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.sha_engine_top.sha256_sigma0_sigma1.sigma_0' is assigned asynchronously"	sha256_msg_datapath.v :71	crypto_alu.sha_engine_top.sha256_sigma0_sigma1-sha256_msg_datapath	-
95	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.sha_engine_top.sha256_sigma0_sigma1.sigma_1' is assigned asynchronously"	sha256_msg_datapath.v :72	crypto_alu.sha_engine_top.sha256_sigma0_sigma1-sha256_msg_datapath	-
96	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no	OTP_NR_ASYA	"Output port 'crypto_alu.sha_engine_top.sha256_sigma0_sigma1.w0_out' is assigned asynchronously"	sha256_msg_datapath.v :73	crypto_alu.sha_engine_top.sha256_sigma0_sigma1-sha256_msg_datapath	-

	functional or synthesis impact.)					
97	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.sha_engine_top.sha256_sigma0_sigma1.w1_out' is assigned asynchronously"	sha256_msg_datapath.v :74	crypto_alu.sha_engine_top.sha256_sigma0_sigma1-sha256_msg_datapath	-
98	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.sha_engine_top.sha256_sigma0_sigma1.w2_out' is assigned asynchronously"	sha256_msg_datapath.v :75	crypto_alu.sha_engine_top.sha256_sigma0_sigma1-sha256_msg_datapath	-
99	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.sha_engine_top.sha256_sigma0_sigma1.w3_out' is assigned asynchronously"	sha256_msg_datapath.v :76	crypto_alu.sha_engine_top.sha256_sigma0_sigma1-sha256_msg_datapath	-
100	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.sha_engine_top.sha_engine_out' is assigned asynchronously"	sha_engine.v :3	crypto_alu.sha_engine_top-sha_engine	-
101	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.aes_out' is assigned asynchronously"	crypto_engine.v :10	crypto_alu-crypto_engine	-
102	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.bitmanipulation_engine_out' is assigned asynchronously"	crypto_engine.v :8	crypto_alu-crypto_engine	-
103	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.carry_less_mul_out' is assigned asynchronously"	crypto_engine.v :9	crypto_alu-crypto_engine	-

104	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'crypto_alu.sha_out' is assigned asynchronously"	crypto_engine.v :11	crypto_alu-crypto_engine	-
105	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'alu_sel_mux.alu_out' is assigned asynchronously"	mux_31.v :6	alu_sel_mux-mux31	-
106	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'dm.d1.dout' is assigned asynchronously"	data_memory.v :6	dm.d1-data_memory	-
107	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'dm.d2.dout' is assigned asynchronously"	data_memory.v :6	dm.d2-data_memory	-
108	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'dm.d3.dout' is assigned asynchronously"	data_memory.v :6	dm.d3-data_memory	-
109	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'dm.d4.dout' is assigned asynchronously"	data_memory.v :6	dm.d4-data_memory	-
110	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure combinational logic and is not required to be registered; no functional or synthesis impact.)	OTP_NR_ASYA	"Output port 'dm.data_out' is assigned asynchronously"	data_memory_top.v :6	dm-data_memory_top	-
111	Waived (OTP_NR_ASYA waived because this output is intentionally driven by pure	OTP_NR_ASYA	"Output port 'data_alu_sel.mux_out' is assigned asynchronously"	mux_21.v :4	data_alu_sel-mux	-

	combinational logic and is not required to be registered; no functional or synthesis impact.)					
112	Waived (OTP_UC_INST waived because the unused upper bits of the carry-less multiplication output are intentionally not required for this operation and have no functional or synthesis impact.)	OTP_UC_INST	"Port 'pc_out[1:0]' (which is being used as an output) of design-unit 'pc_top' is being driven inside the design, but not connected in its instance 'pc_counter'"	riscv_scalar_cryptography_processor.v :29	riscv_scalar_cryptography_proscessor-riscv_scalar_cryptography_proscessor	-
113	Waived (OTP_UC_INST waived because the unused upper bits of the carry-less multiplication output are intentionally not required for this operation and have no functional or synthesis impact.)	OTP_UC_INST	"Port 'pc_out[31:10]' (which is being used as an output) of design-unit 'pc_top' is being driven inside the design, but not connected in its instance 'pc_counter'"	riscv_scalar_cryptography_processor.v :29	riscv_scalar_cryptography_proscessor-riscv_scalar_cryptography_proscessor	-
114	Waived (OTP_UC_INST waived because the unused upper bits of the carry-less multiplication output are intentionally not required for this operation and have no functional or synthesis impact.)	OTP_UC_INST	"Port 'ao' (which is being used as an output) of design-unit 'sha256_round' is being driven inside the design, but not connected in its instance 'crypto_alu.sha_engine_top.sha_sum0_sum1'"	sha_engine.v :12	crypto_alu.sha_engine_top-sha_engine	-
115	Waived (OTP_UC_INST waived because the unused upper bits of the carry-less multiplication output are intentionally not required for this operation and have no functional or synthesis impact.)	OTP_UC_INST	"Port 'bo' (which is being used as an output) of design-unit 'sha256_round' is being driven inside the design, but not connected in its instance 'crypto_alu.sha_engine_top.sha_sum0_sum1'"	sha_engine.v :12	crypto_alu.sha_engine_top-sha_engine	-
116	Waived (OTP_UC_INST waived because the unused upper bits of the carry-less multiplication output are intentionally not required for this operation and have no functional or synthesis impact.)	OTP_UC_INST	"Port 'co' (which is being used as an output) of design-unit 'sha256_round' is being driven inside the design, but not connected in its instance 'crypto_alu.sha_engine_top.sha_sum0_sum1'"	sha_engine.v :12	crypto_alu.sha_engine_top-sha_engine	-
117	Waived (OTP_UC_INST waived because the unused upper bits of the carry-less multiplication output are intentionally not required for this operation and have no functional or synthesis impact.)	OTP_UC_INST	"Port 'd0' (which is being used as an output) of design-unit 'sha256_round' is being driven inside the design, but not connected in its instance 'crypto_alu.sha_engine_top.sha_sum0_sum1'"	sha_engine.v :12	crypto_alu.sha_engine_top-sha_engine	-

118	Waived (OTP_UC_INST waived because the unused upper bits of the carry-less multiplication output are intentionally not required for this operation and have no functional or synthesis impact.)	OTP_UC_INST	"Port 'eo' (which is being used as an output) of design-unit 'sha256_round' is being driven inside the design, but not connected in its instance 'crypto_alu.sha_engine_top.sha_sum0_sum1'"	sha_engine.v:12	crypto_alu.sha_engine_top-sha_engine	-
119	Waived (OTP_UC_INST waived because the unused upper bits of the carry-less multiplication output are intentionally not required for this operation and have no functional or synthesis impact.)	OTP_UC_INST	"Port 'fo' (which is being used as an output) of design-unit 'sha256_round' is being driven inside the design, but not connected in its instance 'crypto_alu.sha_engine_top.sha_sum0_sum1'"	sha_engine.v:12	crypto_alu.sha_engine_top-sha_engine	-
120	Waived (OTP_UC_INST waived because the unused upper bits of the carry-less multiplication output are intentionally not required for this operation and have no functional or synthesis impact.)	OTP_UC_INST	"Port 'go' (which is being used as an output) of design-unit 'sha256_round' is being driven inside the design, but not connected in its instance 'crypto_alu.sha_engine_top.sha_sum0_sum1'"	sha_engine.v:12	crypto_alu.sha_engine_top-sha_engine	-
121	Waived (OTP_UC_INST waived because the unused upper bits of the carry-less multiplication output are intentionally not required for this operation and have no functional or synthesis impact.)	OTP_UC_INST	"Port 'ho' (which is being used as an output) of design-unit 'sha256_round' is being driven inside the design, but not connected in its instance 'crypto_alu.sha_engine_top.sha_sum0_sum1'"	sha_engine.v:12	crypto_alu.sha_engine_top-sha_engine	-
122	Waived (OTP_UC_INST waived because the unused upper bits of the carry-less multiplication output are intentionally not required for this operation and have no functional or synthesis impact.)	OTP_UC_INST	"Port 'w0_out' (which is being used as an output) of design-unit 'sha256_msg_datapath' is being driven inside the design, but not connected in its instance 'crypto_alu.sha_engine_top.sha256_sigma0_sigma1'"	sha_engine.v:35	crypto_alu.sha_engine_top-sha_engine	-
123	Waived (OTP_UC_INST waived because the unused upper bits of the carry-less multiplication output are intentionally not required for this operation and have no functional or synthesis impact.)	OTP_UC_INST	"Port 'w1_out' (which is being used as an output) of design-unit 'sha256_msg_datapath' is being driven inside the design, but not connected in its instance 'crypto_alu.sha_engine_top.sha256_sigma0_sigma1'"	sha_engine.v:35	crypto_alu.sha_engine_top-sha_engine	-
124	Waived (OTP_UC_INST waived because the unused upper bits of the carry-less multiplication	OTP_UC_INST	"Port 'w2_out' (which is being used as an output) of design-unit 'sha256_msg_datapath' is being driven inside the design, but not connected in its instance 'crypto_alu.sha_engine_top.sha256_sigma0_sigma1'"	sha_engine.v:35	crypto_alu.sha_engine_top-sha_engine	-

	output are intentionally not required for this operation and have no functional or synthesis impact.)					
125	Waived (OTP_UC_INST waived because the unused upper bits of the carry-less multiplication output are intentionally not required for this operation and have no functional or synthesis impact.)	OTP_UC_INST	"Port 'w3_out' (which is being used as an output) of design-unit 'sha256_msg_datapath' is being driven inside the design, but not connected in its instance 'crypto_alu.sha_engine_top.sha256_sigma0_sigma1'"	sha_engine.v :35	crypto_alu.sha_engine_top-sha_engine	-
126	Waived (Wire is not used in logical expression. it only used for connect)	WIR_NO_READ	"Wire 'pc_in_wide[32]' declared in module 'pc_add' does not drive any object, but is assigned at least once"	pc_adder.v :16	*-pc_add	-
127	Waived (Wire is not used in logical expression. it only used for connect)	WIR_NO_READ	"Wire 'lshift_wide[32]' declared in module 'bit_ror' does not drive any object, but is assigned at least once"	bit_ror.v :51	*-bit_ror	-
128	Waived (Wire is not used in logical expression. it only used for connect)	WIR_NO_READ	"Wire 'T1_wide[34:32]' declared in module 'sha256_round' does not drive any object, but is assigned at least once"	sha256_round.v :92	*-sha256_round	-
129	Waived (Wire is not used in logical expression. it only used for connect)	WIR_NO_READ	"Wire 'T2_wide[33:32]' declared in module 'sha256_round' does not drive any object, but is assigned at least once"	sha256_round.v :93	*-sha256_round	-
130	Waived (Wire is not used in logical expression. it only used for connect)	WIR_NO_READ	"Wire 'ao_wide[32]' declared in module 'sha256_round' does not drive any object, but is assigned at least once"	sha256_round.v :94	*-sha256_round	-
131	Waived (Wire is not used in logical expression. it only used for connect)	WIR_NO_READ	"Wire 'eo_wide[32]' declared in module 'sha256_round' does not drive any object, but is assigned at least once"	sha256_round.v :95	*-sha256_round	-
132	Waived (Wire is not used in logical expression. it only used for connect)	WIR_NO_READ	"Wire 'w0_wide[33:32]' declared in module 'sha256_msg_datapath' does not drive any object, but is assigned at least once"	sha256_msg_datapath.v :132	*-sha256_msg_datapath	-
133	Waived (Wire is not used in logical expression. it only used for connect)	WIR_NO_READ	"Wire 'w1_wide[33:32]' declared in module 'sha256_msg_datapath' does not drive any object, but is assigned at least once"	sha256_msg_datapath.v :133	*-sha256_msg_datapath	-
134	Waived (Wire is not used in logical expression. it only used for connect)	WIR_NO_READ	"Wire 'w2_wide[33:32]' declared in module 'sha256_msg_datapath' does not drive any object, but is assigned at least once"	sha256_msg_datapath.v :134	*-sha256_msg_datapath	-
135	Waived (Wire is not used in logical expression. it only used for connect)	WIR_NO_READ	"Wire 'w3_wide[33:32]' declared in module 'sha256_msg_datapath' does not drive any object, but is assigned at least once"	sha256_msg_datapath.v :135	*-sha256_msg_datapath	-
136	Waived (Wire is not used in logical expression. it only used for connect)	WIR_NO_READ	"Wire 'unused_a' declared in module 'sha_engine' does not drive any object, but is assigned at least once"	sha_engine.v :7	*-sha_engine	-
137	Waived (Wire is not used in logical expression. it only used for connect)	WIR_NO_READ	"Wire 'unused_b' declared in module 'sha_engine' does not drive any object, but is assigned at least once"	sha_engine.v :7	*-sha_engine	-

138	Waived (Wire is not used in logical expression. it only used for connect)	WIR_NO_READ	"Wire 'unused_c' declared in module 'sha_engine' does not drive any object, but is assigned at least once"	sha_engine.v :7	*-sha_engine	-
139	Waived (Wire is not used in logical expression. it only used for connect)	WIR_NO_READ	"Wire 'unused_d' declared in module 'sha_engine' does not drive any object, but is assigned at least once"	sha_engine.v :7	*-sha_engine	-
140	Waived (Wire is not used in logical expression. it only used for connect)	WIR_NO_READ	"Wire 'unused_e' declared in module 'sha_engine' does not drive any object, but is assigned at least once"	sha_engine.v :7	*-sha_engine	-
141	Waived (Wire is not used in logical expression. it only used for connect)	WIR_NO_READ	"Wire 'unused_f' declared in module 'sha_engine' does not drive any object, but is assigned at least once"	sha_engine.v :7	*-sha_engine	-
142	Waived (Wire is not used in logical expression. it only used for connect)	WIR_NO_READ	"Wire 'unused_g' declared in module 'sha_engine' does not drive any object, but is assigned at least once"	sha_engine.v :7	*-sha_engine	-
143	Waived (Wire is not used in logical expression. it only used for connect)	WIR_NO_READ	"Wire 'unused_h' declared in module 'sha_engine' does not drive any object, but is assigned at least once"	sha_engine.v :7	*-sha_engine	-
144	Waived (Wire is not used in logical expression. it only used for connect)	WIR_NO_READ	"Wire 'unused_i' declared in module 'sha_engine' does not drive any object, but is assigned at least once"	sha_engine.v :8	*-sha_engine	-
145	Waived (Wire is not used in logical expression. it only used for connect)	WIR_NO_READ	"Wire 'unused_j' declared in module 'sha_engine' does not drive any object, but is assigned at least once"	sha_engine.v :8	*-sha_engine	-
146	Waived (Wire is not used in logical expression. it only used for connect)	WIR_NO_READ	"Wire 'unused_k' declared in module 'sha_engine' does not drive any object, but is assigned at least once"	sha_engine.v :8	*-sha_engine	-
147	Waived (Wire is not used in logical expression. it only used for connect)	WIR_NO_READ	"Wire 'unused_l' declared in module 'sha_engine' does not drive any object, but is assigned at least once"	sha_engine.v :8	*-sha_engine	-
148	Waived (Wire is not used in logical expression. it only used for connect)	WIR_NO_READ	"Wire 'pc_inst_wire[1:0]' declared in module 'riscv_scalar_cryptography_proscessor' does not drive any object, but is assigned at least once"	riscv_scalar_cryptography_processor.v :11	*-riscv_scalar_cryptography_proscessor	-
149	Waived (Wire is not used in logical expression. it only used for connect)	WIR_NO_READ	"Wire 'pc_inst_wire[31:10]' declared in module 'riscv_scalar_cryptography_proscessor' does not drive any object, but is assigned at least once"	riscv_scalar_cryptography_processor.v :11	*-riscv_scalar_cryptography_proscessor	-

STRUCTURAL

#	Severity	Tag	Message	Source Location	Instance-Module	Grouping Status
1	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on	INS_NR_INPR	"Input port 'pc_counter.pc_adder.pc_next' of instance 'pc_counter.pc_adder' is not a register"	pc_adder.v :12	pc_counter.pc_adder-pc_add	-

	synthesis or functionality)					
2	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'instruction.im_memo0.pc_addr' of instance 'instruction.im_memo0' is not a register"	im_mem0.v :3	instruction.im_memo0-im_mem0	-
3	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'instruction.im_memo0.write_data' of instance 'instruction.im_memo0' is not a register"	im_mem0.v :6	instruction.im_memo0-im_mem0	-
4	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'instruction.im_memo1.pc_addr' of instance 'instruction.im_memo1' is not a register"	im_mem1.v :3	instruction.im_memo1-im_mem1	-
5	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'instruction.im_memo1.write_data' of instance 'instruction.im_memo1' is not a register"	im_mem1.v :6	instruction.im_memo1-im_mem1	-
6	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No	INS_NR_INPR	"Input port 'instruction.im_memo2.pc_addr' of instance 'instruction.im_memo2' is not a register"	im_mem2.v :3	instruction.im_memo2-im_mem2	-

	impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)					
7	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'instruction.im_memo2.write_data' of instance 'instruction.im_memo2' is not a register"	im_mem2.v :6	instruction.im_memo2-im_mem2	-
8	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'instruction.im_memo3.pc_addr' of instance 'instruction.im_memo3' is not a register"	im_mem3.v :2	instruction.im_memo3-im_mem3	-
9	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'instruction.im_memo3.write_data' of instance 'instruction.im_memo3' is not a register"	im_mem3.v :5	instruction.im_memo3-im_mem3	-
10	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'instruction.pc_addr' of instance 'instruction' is not a register"	im_mem_top.v :2	instruction-instruction_memory_top	-
11	Waived (Pure combinational logic by design. Output is intended to	INS_NR_INPR	"Input port 'instruction.write_data' of instance 'instruction' is not a register"	im_mem_top.v :5	instruction-instruction_memory_top	-

	change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)					
12	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'riscv_contro_unit.func3' of instance 'riscv_contro_unit' is not a register"	riscv_control_unit.v :3	riscv_contro_unit-riscv_control_unit	-
13	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'riscv_contro_unit.func7' of instance 'riscv_contro_unit' is not a register"	riscv_control_unit.v :4	riscv_contro_unit-riscv_control_unit	-
14	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'riscv_contro_unit.inst_in[31:20]' of instance 'riscv_contro_unit' is not a register"	riscv_control_unit.v :6	riscv_contro_unit-riscv_control_unit	-
15	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on	INS_NR_INPR	"Input port 'riscv_contro_unit.opcode' of instance 'riscv_contro_unit' is not a register"	riscv_control_unit.v :5	riscv_contro_unit-riscv_control_unit	-

	synthesis or functionality)					
16	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'register_file.rd_addr' of instance 'register_file' is not a register"	register_file.v :5	register_file-gpr	-
17	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'register_file.reg_write' of instance 'register_file' is not a register"	register_file.v :6	register_file-gpr	-
18	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'register_file.rs1_addr' of instance 'register_file' is not a register"	register_file.v :3	register_file-gpr	-
19	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'register_file.rs2_addr' of instance 'register_file' is not a register"	register_file.v :4	register_file-gpr	-
20	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No	INS_NR_INPR	"Input port 'register_file.write_data' of instance 'register_file' is not a register"	register_file.v :7	register_file-gpr	-

	impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)					
21	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'signextend.imm' of instance 'signextend' is not a register"	sign_extend.v :1	signextend-sign_extend	-
22	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'signextend.imm_extend' of instance 'signextend' is not a register"	sign_extend.v :2	signextend-sign_extend	-
23	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'mux21.mux_sel' of instance 'mux21' is not a register"	mux_21.v :3	mux21-mux	-
24	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'mux21.rs1' of instance 'mux21' is not a register"	mux_21.v :1	mux21-mux	-
25	Waived (Pure combinational logic by design. Output is intended to	INS_NR_INPR	"Input port 'mux21.rs2' of instance 'mux21' is not a register"	mux_21.v :2	mux21-mux	-

	change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)					
26	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.bit_manipulation_engine.bitrol.rs1' of instance 'crypto_alu.bit_manipulation_engine.bitrol' is not a register"	bit_rol.v :23	crypto_alu.bit_manipulation_engine.bitrol-bit_rol	-
27	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.bit_manipulation_engine.bitrol.rs2[4:0]' of instance 'crypto_alu.bit_manipulation_engine.bitrol' is not a register"	bit_rol.v :24	crypto_alu.bit_manipulation_engine.bitrol-bit_rol	-
28	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.bit_manipulation_engine.bitror.rs1' of instance 'crypto_alu.bit_manipulation_engine.bitror' is not a register"	bit_ror.v :42	crypto_alu.bit_manipulation_engine.bitror-bit_ror	-
29	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on	INS_NR_INPR	"Input port 'crypto_alu.bit_manipulation_engine.bitror.rs2[4:0]' of instance 'crypto_alu.bit_manipulation_engine.bitror' is not a register"	bit_ror.v :43	crypto_alu.bit_manipulation_engine.bitror-bit_ror	-

	synthesis or functionality)					
30	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.bit_manipulation_engine.bitandn.rs1' of instance 'crypto_alu.bit_manipulation_engine.bitandn' is not a register"	bit_andn.v :2	crypto_alu.bit_manipulation_engine.bitandn-bit_andn	-
31	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.bit_manipulation_engine.bitandn.rs2' of instance 'crypto_alu.bit_manipulation_engine.bitandn' is not a register"	bit_andn.v :3	crypto_alu.bit_manipulation_engine.bitandn-bit_andn	-
32	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.bit_manipulation_engine.bitorn.rs1' of instance 'crypto_alu.bit_manipulation_engine.bitorn' is not a register"	bit_orn.v :2	crypto_alu.bit_manipulation_engine.bitorn-bit_orn	-
33	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.bit_manipulation_engine.bitorn.rs2' of instance 'crypto_alu.bit_manipulation_engine.bitorn' is not a register"	bit_orn.v :3	crypto_alu.bit_manipulation_engine.bitorn-bit_orn	-
34	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No	INS_NR_INPR	"Input port 'crypto_alu.bit_manipulation_engine.bitxnor.rs1' of instance 'crypto_alu.bit_manipulation_engine.bitxnor' is not a register"	bit_xnor.v :2	crypto_alu.bit_manipulation_engine.bitxnor-bit_xnor	-

	impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)					
35	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.bit_manipulation_engine.bitxnor.rs2' of instance 'crypto_alu.bit_manipulation_engine.bitxnor' is not a register"	bit_xnor.v :3	crypto_alu.bit_manipulation_engine.bitxnor-bit_xnor	-
36	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.bit_manipulation_engine.bitpack.rs1[15:0]' of instance 'crypto_alu.bit_manipulation_engine.bitpack' is not a register"	bit_pack.v :2	crypto_alu.bit_manipulation_engine.bitpack-bit_pack	-
37	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.bit_manipulation_engine.bitpack.rs2[31:16]' of instance 'crypto_alu.bit_manipulation_engine.bitpack' is not a register"	bit_pack.v :3	crypto_alu.bit_manipulation_engine.bitpack-bit_pack	-
38	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.bit_manipulation_engine.bitpackh.rs1[7:0]' of instance 'crypto_alu.bit_manipulation_engine.bitpackh' is not a register"	bit_packh.v :2	crypto_alu.bit_manipulation_engine.bitpackh-bit_packh	-
39	Waived (Pure combinational logic by design. Output is intended to	INS_NR_INPR	"Input port 'crypto_alu.bit_manipulation_engine.bitpackh.rs2[7:0]' of instance 'crypto_alu.bit_manipulation_engine.bitpackh' is not a register"	bit_packh.v :3	crypto_alu.bit_manipulation_engine.bitpackh-bit_packh	-

	change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)					
40	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.bit_manipulation_engine.bitzip.rs1' of instance 'crypto_alu.bit_manipulation_engine.bitzip' is not a register"	bit_zip.v :17	crypto_alu.bit_manipulation_engine.bitzip-bit_zip	-
41	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.bit_manipulation_engine.bitunzip.rs1' of instance 'crypto_alu.bit_manipulation_engine.bitunzip' is not a register"	bit_unzip.v :18	crypto_alu.bit_manipulation_engine.bitunzip-bit_unzip	-
42	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.bit_manipulation_engine.bitbrev.rs1' of instance 'crypto_alu.bit_manipulation_engine.bitbrev' is not a register"	bit_brev.v :2	crypto_alu.bit_manipulation_engine.bitbrev-bit_brev8	-
43	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on	INS_NR_INPR	"Input port 'crypto_alu.bit_manipulation_engine.bitrev8.rs1' of instance 'crypto_alu.bit_manipulation_engine.bitrev8' is not a register"	bit_rev8.v :2	crypto_alu.bit_manipulation_engine.bitrev8-bit_rev8	-

	synthesis or functionality)					
44	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.bit_manipulation_engine.alu_opcode' of instance 'crypto_alu.bit_manipulation_engine' is not a register"	bitmanipulation.v :4	crypto_alu.bit_manipulation_engine-bitmanipulation	-
45	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.bit_manipulation_engine.rs1' of instance 'crypto_alu.bit_manipulation_engine' is not a register"	bitmanipulation.v :2	crypto_alu.bit_manipulation_engine-bitmanipulation	-
46	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.bit_manipulation_engine.rs2' of instance 'crypto_alu.bit_manipulation_engine' is not a register"	bitmanipulation.v :3	crypto_alu.bit_manipulation_engine-bitmanipulation	-
47	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.carry_less_mul_top.carry_less_mul_lower_32.x' of instance 'crypto_alu.carry_less_mul_top.carry_less_mul_lower_32' is not a register"	carry_less_multiplication.v :83	crypto_alu.carry_less_mul_top.carry_less_mul_lower_32-carry_less_mul	-
48	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No	INS_NR_INPR	"Input port 'crypto_alu.carry_less_mul_top.carry_less_mul_lower_32.y' of instance 'crypto_alu.carry_less_mul_top.carry_less_mul_lower_32' is not a register"	carry_less_multiplication.v :84	crypto_alu.carry_less_mul_top.carry_less_mul_lower_32-carry_less_mul	-

	impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)					
49	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.carry_less_mul_top.alu_opcode' of instance 'crypto_alu.carry_less_mul_top' is not a register"	carry_less_engine.v :4	crypto_alu.carry_less_mul_top-carry_less_mul_top	-
50	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.carry_less_mul_top.rs1' of instance 'crypto_alu.carry_less_mul_top' is not a register"	carry_less_engine.v :2	crypto_alu.carry_less_mul_top-carry_less_mul_top	-
51	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.carry_less_mul_top.rs2' of instance 'crypto_alu.carry_less_mul_top' is not a register"	carry_less_engine.v :3	crypto_alu.carry_less_mul_top-carry_less_mul_top	-
52	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.aes_engine_top.aes32sbox.sbox.in_byte' of instance 'crypto_alu.aes_engine_top.aes32sbox.sbox' is not a register"	sbox_values.v :2	crypto_alu.aes_engine_top.aes32sbox.sbox-aes_sbox	-
53	Waived (Pure combinational logic by design. Output is intended to	INS_NR_INPR	"Input port 'crypto_alu.aes_engine_top.aes32sbox.byte_sel' of instance 'crypto_alu.aes_engine_top.aes32sbox' is not a register"	aes32esi.v :4	crypto_alu.aes_engine_top.aes32sbox-aes32esi	-

	change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)					
54	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.aes_engine_top.aes32sbox.bytes' of instance 'crypto_alu.aes_engine_top.aes32sbox' is not a register"	aes32esi.v :5	crypto_alu.aes_engine_top.aes32sbox-aes32esi	-
55	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.aes_engine_top.aes32sbox.rs1' of instance 'crypto_alu.aes_engine_top.aes32sbox' is not a register"	aes32esi.v :2	crypto_alu.aes_engine_top.aes32sbox-aes32esi	-
56	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.aes_engine_top.aes32sbox.rs2' of instance 'crypto_alu.aes_engine_top.aes32sbox' is not a register"	aes32esi.v :3	crypto_alu.aes_engine_top.aes32sbox-aes32esi	-
57	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on	INS_NR_INPR	"Input port 'crypto_alu.aes_engine_top.aes32smix.sbox.in_byte' of instance 'crypto_alu.aes_engine_top.aes32smix.sbox' is not a register"	sbox_values.v :2	crypto_alu.aes_engine_top.aes32smix.sbox-aes_sbox	-

	synthesis or functionality)					
58	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.aes_engine_top.aes32smix.mixcolumn.in_word[31:24]' of instance 'crypto_alu.aes_engine_top.aes32smix.mixcolumn' is not a register"	mix_op.v :63	crypto_alu.aes_engine_top.aes32smix.mixcolumn-mix_op	-
59	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.aes_engine_top.aes32smix.byte_sel' of instance 'crypto_alu.aes_engine_top.aes32smix' is not a register"	aes32esmi.v :4	crypto_alu.aes_engine_top.aes32smix-aes32esmi	-
60	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.aes_engine_top.aes32smix.bytes' of instance 'crypto_alu.aes_engine_top.aes32smix' is not a register"	aes32esmi.v :5	crypto_alu.aes_engine_top.aes32smix-aes32esmi	-
61	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.aes_engine_top.aes32smix.rs1' of instance 'crypto_alu.aes_engine_top.aes32smix' is not a register"	aes32esmi.v :2	crypto_alu.aes_engine_top.aes32smix-aes32esmi	-
62	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No	INS_NR_INPR	"Input port 'crypto_alu.aes_engine_top.aes32smix.rs2' of instance 'crypto_alu.aes_engine_top.aes32smix' is not a register"	aes32esmi.v :3	crypto_alu.aes_engine_top.aes32smix-aes32esmi	-

	impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)					
63	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.aes_engine_top.alu_opcode' of instance 'crypto_alu.aes_engine_top' is not a register"	aes_engine.v :6	crypto_alu.aes_engine_top-aes_engine	-
64	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.aes_engine_top.byte_sel' of instance 'crypto_alu.aes_engine_top' is not a register"	aes_engine.v :5	crypto_alu.aes_engine_top-aes_engine	-
65	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.aes_engine_top.bytes' of instance 'crypto_alu.aes_engine_top' is not a register"	aes_engine.v :4	crypto_alu.aes_engine_top-aes_engine	-
66	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.aes_engine_top.rs1' of instance 'crypto_alu.aes_engine_top' is not a register"	aes_engine.v :2	crypto_alu.aes_engine_top-aes_engine	-
67	Waived (Pure combinational logic by design. Output is intended to	INS_NR_INPR	"Input port 'crypto_alu.aes_engine_top.rs2' of instance 'crypto_alu.aes_engine_top' is not a register"	aes_engine.v :3	crypto_alu.aes_engine_top-aes_engine	-

	change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)					
68	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.sha_engine_top.sha_sum0_sum1.a' of instance 'crypto_alu.sha_engine_top.sha_sum0_sum1' is not a register"	sha256_round.v :52	crypto_alu.sha_engine_top.sha_sum0_sum1-sha256_round	-
69	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.sha_engine_top.sha256_sigma0_sigma1.bufrd_t16_in' of instance 'crypto_alu.sha_engine_top.sha256_sigma0_sigma1' is not a register"	sha256_msg_datapath.v :59	crypto_alu.sha_engine_top.sha256_sigma0_sigma1-sha256_msg_datapath	-
70	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.sha_engine_top.alu_opcode' of instance 'crypto_alu.sha_engine_top' is not a register"	sha_engine.v :2	crypto_alu.sha_engine_top-sha_engine	-
71	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on	INS_NR_INPR	"Input port 'crypto_alu.sha_engine_top.rs1' of instance 'crypto_alu.sha_engine_top' is not a register"	sha_engine.v :1	crypto_alu.sha_engine_top-sha_engine	-

	synthesis or functionality)					
72	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.alu_opcode' of instance 'crypto_alu' is not a register"	crypto_engine.v :5	crypto_alu-crypto_engine	-
73	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.byte_sel' of instance 'crypto_alu' is not a register"	crypto_engine.v :7	crypto_alu-crypto_engine	-
74	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.bytes' of instance 'crypto_alu' is not a register"	crypto_engine.v :6	crypto_alu-crypto_engine	-
75	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'crypto_alu.rs1' of instance 'crypto_alu' is not a register"	crypto_engine.v :3	crypto_alu-crypto_engine	-
76	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No	INS_NR_INPR	"Input port 'crypto_alu.rs2' of instance 'crypto_alu' is not a register"	crypto_engine.v :4	crypto_alu-crypto_engine	-

	impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)					
77	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'alu_sel_mux.aes_in' of instance 'alu_sel_mux' is not a register"	mux_31.v:3	alu_sel_mux-mux31	-
78	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'alu_sel_mux.bit_mani_in' of instance 'alu_sel_mux' is not a register"	mux_31.v:1	alu_sel_mux-mux31	-
79	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'alu_sel_mux.carry_less_in' of instance 'alu_sel_mux' is not a register"	mux_31.v:2	alu_sel_mux-mux31	-
80	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'alu_sel_mux.mux_sel_alu' of instance 'alu_sel_mux' is not a register"	mux_31.v:5	alu_sel_mux-mux31	-
81	Waived (Pure combinational logic by design. Output is intended to	INS_NR_INPR	"Input port 'alu_sel_mux.sha_in' of instance 'alu_sel_mux' is not a register"	mux_31.v:4	alu_sel_mux-mux31	-

	change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)					
82	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'dm.d1.addr' of instance 'dm.d1' is not a register"	data_memory.v :2	dm.d1-data_memory	-
83	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'dm.d1.write_data' of instance 'dm.d1' is not a register"	data_memory.v :5	dm.d1-data_memory	-
84	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'dm.d2.addr' of instance 'dm.d2' is not a register"	data_memory.v :2	dm.d2-data_memory	-
85	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on	INS_NR_INPR	"Input port 'dm.d2.write_data' of instance 'dm.d2' is not a register"	data_memory.v :5	dm.d2-data_memory	-

	synthesis or functionality)					
86	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'dm.d3.addr' of instance 'dm.d3' is not a register"	data_memory.v :2	dm.d3-data_memory	-
87	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'dm.d3.write_data' of instance 'dm.d3' is not a register"	data_memory.v :5	dm.d3-data_memory	-
88	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'dm.d4.addr' of instance 'dm.d4' is not a register"	data_memory.v :2	dm.d4-data_memory	-
89	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'dm.d4.write_data' of instance 'dm.d4' is not a register"	data_memory.v :5	dm.d4-data_memory	-
90	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No	INS_NR_INPR	"Input port 'dm.addr[9:2]' of instance 'dm' is not a register"	data_memory_top.v :1	dm-data_memory_top	-

	impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)					
91	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'dm.data_write_en' of instance 'dm' is not a register"	data_memory_top.v :5	dm-data_memory_top	-
92	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'dm.write_data' of instance 'dm' is not a register"	data_memory_top.v :4	dm-data_memory_top	-
93	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'data_alu_sel.mux_sel' of instance 'data_alu_sel' is not a register"	mux_21.v :3	data_alu_sel-mux	-
94	Waived (Pure combinational logic by design. Output is intended to change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)	INS_NR_INPR	"Input port 'data_alu_sel.rs1' of instance 'data_alu_sel' is not a register"	mux_21.v :1	data_alu_sel-mux	-
95	Waived (Pure combinational logic by design. Output is intended to	INS_NR_INPR	"Input port 'data_alu_sel.rs2' of instance 'data_alu_sel' is not a register"	mux_21.v :2	data_alu_sel-mux	-

	change immediately with input changes.No state or storage element is required. No impact on synthesis or functionality.No state or storage element is required. No impact on synthesis or functionality)					
96	Waived (Module intentionally uses synchronous write and asynchronous read memory behavior. Standard memory modeling style with no synthesis or functional impact)	MOD_IS_SYAS	"The design-unit 'data_memory' contains synchronous as well as asynchronous logic. Asynchronous logic is present at 'dout[7]' and Synchronous logic at 'mem[145][7]'"	data_memory.v :9	*-data_memory	-
97	Waived (Module intentionally uses synchronous write and asynchronous read memory behavior. Standard memory modeling style with no synthesis or functional impact)	MOD_IS_SYAS	"The design-unit 'gpr' contains synchronous as well as asynchronous logic. Asynchronous logic is present at 'rs2_data[31]' and Synchronous logic at 'mem[2][31]'"	register_file.v :12	*-gpr	-
98	Waived (Module intentionally uses synchronous write and asynchronous read memory behavior. Standard memory modeling style with no synthesis or functional impact)	MOD_IS_SYAS	"The design-unit 'im_mem0' contains synchronous as well as asynchronous logic. Asynchronous logic is present at 'byte_0_out[7]' and Synchronous logic at 'mem[55][7]'"	im_mem0.v :11	*-im_mem0	-
99	Waived (Module intentionally uses synchronous write and asynchronous read memory behavior. Standard memory modeling style with no synthesis or functional impact)	MOD_IS_SYAS	"The design-unit 'im_mem1' contains synchronous as well as asynchronous logic. Asynchronous logic is present at 'byte_1_out[7]' and Synchronous logic at 'mem[11][7]'"	im_mem1.v :12	*-im_mem1	-
100	Waived (Module intentionally uses synchronous write and asynchronous read memory behavior. Standard memory modeling style with no synthesis or functional impact)	MOD_IS_SYAS	"The design-unit 'im_mem2' contains synchronous as well as asynchronous logic. Asynchronous logic is present at 'byte_2_out[7]' and Synchronous logic at 'mem[210][7]'"	im_mem2.v :11	*-im_mem2	-

101	Waived (Module intentionally uses synchronous write and asynchronous read memory behavior. Standard memory modeling style with no synthesis or functional impact)	MOD_IS_SYAS	"The design-unit 'im_mem3' contains synchronous as well as asynchronous logic. Asynchronous logic is present at 'byte_3_out[7]' and Synchronous logic at 'mem[166][7]'"	im_mem3.v :9	*-im_mem3	-
102	Waived (its is yop module output its not connect to any ff)	MOD_NO_IPRG	"Input port 'inst_write_en' of top-level design-unit is not registered"	riscv_scalar_cryptography_processor.v :4	riscv_scalar_cryptography_proscessor-riscv_scalar_cryptography_processor	-
103	Waived (Yes this is a glue logic . why i am using this because of synthesis to find the critical path)	MOD_NS_GLGC	"Glue logic inferred in top-level design-unit 'riscv_scalar_cryptography_proscessor'"	riscv_scalar_cryptography_processor.v :198	riscv_scalar_cryptography_proscessor-riscv_scalar_cryptography_proscessor	-
104	Waived (NET_NO_LOAD waived because this signal is driven but intentionally not connected to any consuming logic and has no functional impact.)	NET_NO_LOAD	"Net 'pc_counter.pc_adder.pc_in_wide[32]' declared in design-unit 'pc_add' has a driver but has no load"	pc_adder.v :16	pc_counter.pc_adder-pc_add	-
105	Waived (NET_NO_LOAD waived because this signal is driven but intentionally not connected to any consuming logic and has no functional impact.)	NET_NO_LOAD	"Net 'crypto_alu.bit_manipulation_engine.bitror.lshift_wide[32]' declared in design-unit 'bit_ror' has a driver but has no load"	bit_ror.v :51	crypto_alu.bit_manipulation_engine.bitror-bit_ror	-
106	Waived (NET_NO_LOAD waived because this signal is driven but intentionally not connected to any consuming logic and has no functional impact.)	NET_NO_LOAD	"Net 'crypto_alu.sha_engine_top.sha_sum0_sum1.T1_wide[34:32]' declared in design-unit 'sha256_round' has a driver but has no load"	sha256_round.v :92	crypto_alu.sha_engine_top.sha_sum0_sum1-sha256_round	-
107	Waived (NET_NO_LOAD waived because this signal is driven but intentionally not connected to any consuming logic and has no functional impact.)	NET_NO_LOAD	"Net 'crypto_alu.sha_engine_top.sha_sum0_sum1.T2_wide[33:32]' declared in design-unit 'sha256_round' has a driver but has no load"	sha256_round.v :93	crypto_alu.sha_engine_top.sha_sum0_sum1-sha256_round	-
108	Waived (NET_NO_LOAD waived because this signal is driven but intentionally not connected to any consuming logic and has no functional impact.)	NET_NO_LOAD	"Net 'crypto_alu.sha_engine_top.sha_sum0_sum1.ao_wide' declared in design-unit 'sha256_round' has a driver but has no load"	sha256_round.v :94	crypto_alu.sha_engine_top.sha_sum0_sum1-sha256_round	-
109	Waived (NET_NO_LOAD waived because this signal is driven but intentionally not connected	NET_NO_LOAD	"Net 'crypto_alu.sha_engine_top.sha_sum0_sum1.eo_wide' declared in design-unit 'sha256_round' has a driver but has no load"	sha256_round.v :95	crypto_alu.sha_engine_top.sha_sum0_sum1-sha256_round	-

	to any consuming logic and has no functional impact.)					
110	Waived (NET_NO_LOAD waived because this signal is driven but intentionally not connected to any consuming logic and has no functional impact.)	NET_NO_LOAD	"Net 'crypto_alu.sha_engine_top.sha256_sigma0_sigma1.w0_wide[33:32]' declared in design-unit 'sha256_msg_datapath' has a driver but has no load"	sha256_msg_datapath.v:132	crypto_alu.sha_engine_top.sha256_sigma0_sigma1-sha256_msg_datapath	-
111	Waived (NET_NO_LOAD waived because this signal is driven but intentionally not connected to any consuming logic and has no functional impact.)	NET_NO_LOAD	"Net 'crypto_alu.sha_engine_top.sha256_sigma0_sigma1.w1_wide[33:32]' declared in design-unit 'sha256_msg_datapath' has a driver but has no load"	sha256_msg_datapath.v:133	crypto_alu.sha_engine_top.sha256_sigma0_sigma1-sha256_msg_datapath	-
112	Waived (NET_NO_LOAD waived because this signal is driven but intentionally not connected to any consuming logic and has no functional impact.)	NET_NO_LOAD	"Net 'crypto_alu.sha_engine_top.sha256_sigma0_sigma1.w2_int' declared in design-unit 'sha256_msg_datapath' has a driver but has no load"	sha256_msg_datapath.v:140	crypto_alu.sha_engine_top.sha256_sigma0_sigma1-sha256_msg_datapath	-
113	Waived (NET_NO_LOAD waived because this signal is driven but intentionally not connected to any consuming logic and has no functional impact.)	NET_NO_LOAD	"Net 'crypto_alu.sha_engine_top.sha256_sigma0_sigma1.w2_wide[33:32]' declared in design-unit 'sha256_msg_datapath' has a driver but has no load"	sha256_msg_datapath.v:134	crypto_alu.sha_engine_top.sha256_sigma0_sigma1-sha256_msg_datapath	-
114	Waived (NET_NO_LOAD waived because this signal is driven but intentionally not connected to any consuming logic and has no functional impact.)	NET_NO_LOAD	"Net 'crypto_alu.sha_engine_top.sha256_sigma0_sigma1.w3_int' declared in design-unit 'sha256_msg_datapath' has a driver but has no load"	sha256_msg_datapath.v:141	crypto_alu.sha_engine_top.sha256_sigma0_sigma1-sha256_msg_datapath	-
115	Waived (NET_NO_LOAD waived because this signal is driven but intentionally not connected to any consuming logic and has no functional impact.)	NET_NO_LOAD	"Net 'crypto_alu.sha_engine_top.sha256_sigma0_sigma1.w3_wide[33:32]' declared in design-unit 'sha256_msg_datapath' has a driver but has no load"	sha256_msg_datapath.v:135	crypto_alu.sha_engine_top.sha256_sigma0_sigma1-sha256_msg_datapath	-
116	Waived (NET_NO_LOAD waived because this signal is driven but intentionally not connected to any consuming logic and has no functional impact.)	NET_NO_LOAD	"Net 'crypto_alu.sha_engine_top.unused_a' declared in design-unit 'sha_engine' has a driver but has no load"	sha_engine.v:7	crypto_alu.sha_engine_top-sha_engine	-
117	Waived (NET_NO_LOAD waived because this signal is driven but intentionally not connected to any	NET_NO_LOAD	"Net 'crypto_alu.sha_engine_top.unused_b' declared in design-unit 'sha_engine' has a driver but has no load"	sha_engine.v:7	crypto_alu.sha_engine_top-sha_engine	-

	consuming logic and has no functional impact.)					
118	Waived (NET_NO_LOAD waived because this signal is driven but intentionally not connected to any consuming logic and has no functional impact.)	NET_NO_LOAD	"Net 'crypto_alu.sha_engine_top.unused_c' declared in design-unit 'sha_engine' has a driver but has no load"	sha_engine.v :7	crypto_alu.sha_engine_top-sha_engine	-
119	Waived (NET_NO_LOAD waived because this signal is driven but intentionally not connected to any consuming logic and has no functional impact.)	NET_NO_LOAD	"Net 'crypto_alu.sha_engine_top.unused_d' declared in design-unit 'sha_engine' has a driver but has no load"	sha_engine.v :7	crypto_alu.sha_engine_top-sha_engine	-
120	Waived (NET_NO_LOAD waived because this signal is driven but intentionally not connected to any consuming logic and has no functional impact.)	NET_NO_LOAD	"Net 'crypto_alu.sha_engine_top.unused_e' declared in design-unit 'sha_engine' has a driver but has no load"	sha_engine.v :7	crypto_alu.sha_engine_top-sha_engine	-
121	Waived (NET_NO_LOAD waived because this signal is driven but intentionally not connected to any consuming logic and has no functional impact.)	NET_NO_LOAD	"Net 'crypto_alu.sha_engine_top.unused_f' declared in design-unit 'sha_engine' has a driver but has no load"	sha_engine.v :7	crypto_alu.sha_engine_top-sha_engine	-
122	Waived (NET_NO_LOAD waived because this signal is driven but intentionally not connected to any consuming logic and has no functional impact.)	NET_NO_LOAD	"Net 'crypto_alu.sha_engine_top.unused_g' declared in design-unit 'sha_engine' has a driver but has no load"	sha_engine.v :7	crypto_alu.sha_engine_top-sha_engine	-
123	Waived (NET_NO_LOAD waived because this signal is driven but intentionally not connected to any consuming logic and has no functional impact.)	NET_NO_LOAD	"Net 'crypto_alu.sha_engine_top.unused_h' declared in design-unit 'sha_engine' has a driver but has no load"	sha_engine.v :7	crypto_alu.sha_engine_top-sha_engine	-
124	Waived (NET_NO_LOAD waived because this signal is driven but intentionally not connected to any consuming logic and has no functional impact.)	NET_NO_LOAD	"Net 'crypto_alu.sha_engine_top.unused_i' declared in design-unit 'sha_engine' has a driver but has no load"	sha_engine.v :8	crypto_alu.sha_engine_top-sha_engine	-
125	Waived (NET_NO_LOAD waived because this signal is driven but intentionally not connected to any consuming	NET_NO_LOAD	"Net 'crypto_alu.sha_engine_top.unused_j' declared in design-unit 'sha_engine' has a driver but has no load"	sha_engine.v :8	crypto_alu.sha_engine_top-sha_engine	-

	logic and has no functional impact.)					
126	Waived (NET_NO_LOAD waived because this signal is driven but intentionally not connected to any consuming logic and has no functional impact.)	NET_NO_LOAD	"Net 'crypto_alu.sha_engine_top.unused_k' declared in design-unit 'sha_engine' has a driver but has no load"	sha_engine.v :8	crypto_alu.sha_engine_top-sha_engine	-
127	Waived (NET_NO_LOAD waived because this signal is driven but intentionally not connected to any consuming logic and has no functional impact.)	NET_NO_LOAD	"Net 'crypto_alu.sha_engine_top.unused_l' declared in design-unit 'sha_engine' has a driver but has no load"	sha_engine.v :8	crypto_alu.sha_engine_top-sha_engine	-
128	Waived (NET_NO_LOAD waived because this signal is driven but intentionally not connected to any consuming logic and has no functional impact.)	NET_NO_LOAD	"Net 'pc_inst_wire[31:10]' declared in design-unit 'riscv_scalar_cryptography_proscessor' has a driver but has no load"	riscv_scalar_cryptography_processor.v :11	riscv_scalar_cryptography_proscessor-riscv_scalar_cryptography_proscessor	-
129	Waived (NET_NO_LOAD waived because this signal is driven but intentionally not connected to any consuming logic and has no functional impact.)	NET_NO_LOAD	"Net 'pc_inst_wire[1:0]' declared in design-unit 'riscv_scalar_cryptography_proscessor' has a driver but has no load"	riscv_scalar_cryptography_processor.v :11	riscv_scalar_cryptography_proscessor-riscv_scalar_cryptography_proscessor	-

SYNTHESIS

#	Severity	Tag	Message	Source Location	Instance-Module	Grouping Status
1	Waived (resest dont have a constant value)	MOD_NR_ASLD	"In design-unit im_mem0, the reset value of flip-flop mem is not constant. This infers an asynchronous load"	im_mem0.v :15	instruction.im_memo0-im_mem0	-
2	Waived (resest dont have a constant value)	MOD_NR_ASLD	"In design-unit im_mem1, the reset value of flip-flop mem is not constant. This infers an asynchronous load"	im_mem1.v :18	instruction.im_memo1-im_mem1	-
3	Waived (resest dont have a constant value)	MOD_NR_ASLD	"In design-unit im_mem2, the reset value of flip-flop mem is not constant. This infers an asynchronous load"	im_mem2.v :15	instruction.im_memo2-im_mem2	-
4	Waived (resest dont have a constant value)	MOD_NR_ASLD	"In design-unit im_mem3, the reset value of flip-flop mem is not constant. This infers an asynchronous load"	im_mem3.v :13	instruction.im_memo3-im_mem3	-
5	Waived (resest dont have a constant value)	MOD_NR_ASLD	"In design-unit gpr, the reset value of flip-flop mem is not constant. This infers an asynchronous load"	register_file.v :41	register_file-gpr	-
6	Waived (resest dont have a constant value)	MOD_NR_ASLD	"In design-unit data_memory, the reset value of flip-flop mem is not constant. This infers an asynchronous load"	data_memory.v :17	dm.d1-data_memory	-